

Project Report: Banking Customer Analytics Dashboard

1. Objective

Retail banks sit on rich customer data — deposits, loans, savings, credit cards — but it's often scattered across raw tables with no clear narrative. This project takes a raw customer-level banking dataset and turns it into (a) a documented Python EDA process and (b) an interactive Power BI dashboard, with the goal of answering three practical questions:

1. Which customer segments hold the most value (deposits, savings, income)?
2. How are loan and credit exposure distributed across the customer base?
3. Which account types tend to move together, and what does that imply about customer behavior?

2. Dataset Overview

- **Source:** Customer-level retail banking dataset, 3,000 records × 25 columns.
- **Granularity:** One row per customer (`Client ID`).
- **Data quality:** No missing values in any column — a clean starting point that let the project focus on feature engineering and analysis rather than imputation.

Field groups:

- *Demographics:* Age, Nationality, Occupation, Location/Branch ID,

- Gender ID
- *Relationship*: Joined Bank (date), Banking Contact, Loyalty Classification, Fee Structure
- *Financial position*: Estimated Income, Superannuation Savings, Credit Card Balance, Bank Loans, Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Business Lending, Properties Owned
- *Risk*: Risk Weighting

3. Data Cleaning & Feature Engineering

Step	Detail
Date parsing	<code>Joined Bank</code> converted from <code>DD-MM-YYYY</code> string to <code>datetime</code> , enabling tenure-based analysis
Income banding	<code>Estimated Income</code> bucketed into <code>Low</code> ($\leq 100k$), <code>Mid</code> (100k–300k), <code>High</code> (300k+) using <code>pd.cut</code>
Categorical mapping	Gender codes and branch IDs mapped to human-readable labels for the Power BI model
Null check	Confirmed 0 missing values across all 3,000 rows and 25 columns

4. Exploratory Data Analysis

Univariate analysis. Distribution plots (histogram + KDE) were generated for all 11 numeric account/balance fields (Age, Estimated Income, Superannuation Savings, Credit Card Balance, Bank Loans, Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Business Lending, Fee Structure) to check for skew, multi-modality, and outliers before drawing conclusions from summary statistics.

Categorical analysis. Value counts were pulled for every categorical field:

- **Gender:** near-even split (1,512 vs. 1,488)
- **Nationality:** European customers are the largest group (1,309), followed by Asian (754), American (507), Australian (254), and African (176)
- **Loyalty Classification:** skewed toward the base tier — Jade (1,331), Silver (767), Gold (585), Platinum (317)
- **Fee Structure:** High (1,476), Mid (962), Low (562)
- **Risk Weighting:** concentrated at the lower-risk end (levels 1–2 account for the majority of customers)

Correlation analysis. A full correlation matrix was computed across Age, Estimated Income, Superannuation Savings, Credit Card Balance, Bank Loans, Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Business Lending, and Properties Owned, visualized as a heatmap.

Bivariate deep dives. Regression scatter plots were produced for the seven relationships that looked most informative from the heatmap:

- Bank Deposits ↔ Saving Accounts
- Checking Accounts ↔ Saving Accounts
- Checking Accounts ↔ Foreign Currency Account
- Age ↔ Superannuation Savings
- Estimated Income ↔ Checking Accounts
- Bank Loans ↔ Credit Card Balance
- Business Lending ↔ Bank Loans

5. Findings

5.1 Deposit-side accounts move together. Bank Deposits, Checking Accounts, Saving Accounts, and Foreign Currency Account balances are all positively correlated with one another. Practically, this means a customer who holds a high balance in one liquid account tends to hold high balances across the others — they likely represent overlapping measures of a customer's total "funds on hand" rather than independent, siloed products. For the bank, this suggests cross-sell of deposit products should target the same high-liquidity segment rather than being split by account type.

5.2 Income and age track long-term accumulation. Estimated Income and Age show moderate positive correlation with Superannuation Savings, Saving Accounts, and Checking Account balances — consistent with a normal financial lifecycle (older, higher-earning customers have had more time and capacity to accumulate). This is a useful segmentation signal for wealth-management or retirement-product targeting.

5.3 Property ownership is a weak signal. Properties Owned correlates weakly with the banking balance fields. This makes sense — property ownership is driven by factors the dataset doesn't capture (local real estate market, inheritance, geography), so it shouldn't be over-weighted as a proxy for banking wealth.

5.4 Business lending is a distinct customer segment. Business Lending shows a moderate correlation with personal Bank Loans (some customers carry both) but little relationship with deposit-side accounts. This points to a subset of customers whose primary relationship with the bank is credit-driven rather than deposit-driven — a segment that likely needs a different engagement strategy than the deposit-heavy segment in 5.1.

6. Power BI Dashboard

The findings above were operationalized into a 4-page interactive dashboard:

1. **Home** — portfolio KPIs (customer count, total deposits, average balances) for a management-level snapshot.
2. **Loan Analysis** — loan and credit exposure sliced by branch, occupation, and risk weighting.
3. **Deposit Analysis** — deposit/savings breakdown by income band and loyalty tier, surfacing the deposit-correlation pattern from 5.1 visually.
4. **Summary** — consolidated view tying the EDA insights to demographic segments for a non-technical audience.

Income banding and label mapping done in the Python cleaning step

were carried into Power BI as conditional/calculated columns to keep the dashboard model consistent with the EDA.

7. Limitations & Future Work

- The dataset has no time-series/transaction-level detail, so behavior is analyzed at a point-in-time snapshot rather than as a trend.
- Correlation was used to identify relationships; no causal or predictive model was built. A natural next step is a churn or high-value-customer classification model using these features.
- Segment definitions (Income Band, Loyalty tier) are provided/derived directly rather than validated against outcome data (e.g. actual customer profitability).

8. Conclusion

The deposit-side accounts (Deposits, Checking, Savings, Foreign Currency) form one coherent behavioral cluster, income/age explain a moderate share of long-term accumulation, and business lending customers form a separate segment worth targeting differently from the deposit-focused base. These findings are now directly explorable by non-technical stakeholders through the 4-page Power BI dashboard.